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PROJECT TITLE : Smart Accessibility Guide for Sustainable Tourism
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1. **AI Tool(s) Used (Check that all that apply)**

- [☐] None (I/We did not use any AI tools for this assignment)
 [☒] ChatGPT (Version: __5.5__) [☐] Deepseek
 [☐] Gemini [☐] Other: _____

2. **Nature of Assistance (Check all that apply)**

- [☒] **Brainstorming:** Generating ideas, topics, or outlines.
 [☐] **Research:** Summarising long articles or finding concepts.
 [☒] **Editing:** Checking grammar, spelling, or sentence structure.
 [☒] **Coding/Math:** Debugging code or explaining a formula.
 [☐] **Creation:** Generating images, data, or draft text

3. **Disclosure Statement**

Provide a brief explanation of how AI tools were utilised in completing the task and describe the measures taken to ensure that the final submission reflects your own or your group independent work

ChatGPT was used as a supporting tool during the preparation of this task. It assisted in brainstorming project ideas, refining module structures, enhancing grammar and sentence clarity, and providing more deeper explanations for technical concepts and implemental approaches. It was also used to clarify programming-related questions and suggest suitable technologies to use for this proposed system. All AI generated suggestions were reviewed, evaluated and modified by the group to ensure they aligned with the project requirements and reflected our own understanding and decisions. The final project proposal, including module design, and documentation was independently reviewed and revised by the group to ensure its originality and accuracy for the assignment.

4. **Verification & Integrity**

- [☒] I/We have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
 [☒] I/We have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
 [☒] I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own

Student / Group Representative Signature:



Date: 27/6/2026



PROJECT PROPOSAL

PROJECT TITLE :

SMART **A**ccessibility **G**uide for **S**ustainable
Tourism

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1. Introduction

1.1 Project Description

The proposed project will develop a mobile-based Smart Accessibility Guide for Sustainable Tourism that helps disabled and elderly tourists plan, navigate, and enjoy tourist attractions more independently and safely. The application aims to support Sustainable Development Goal (SDG) 11: Sustainable Cities and Communities by promoting inclusive and accessible tourism experiences. Users can create accounts, specify their accessibility needs, discover accessible attractions through advanced search and AI-powered recommendations, plan routes with accessibility considerations, and obtain real-time accessibility information through on-site QR code scanning.

The system integrates several key modules, including user management, accessible attraction directory and AI recommendations, smart route and navigation, tourism support and emergency, and an on-site QR module. Users can search and filter attractions based on accessibility features, receive personalized recommendations, access nearby hospitals and emergency contacts, and share their location during emergencies. The targeted users of the system are wheelchair users, elderly users, and hearing-impaired individuals, while administrators can manage attraction information, moderate reports, and monitor platform activities to ensure accurate and latest accessibility information.

1.2 UN SDG and Justification

- [☐] Goal 8: Decent Work and Economic Growth
- [☒] Goal 11: Sustainable Cities and Communities
- [☐] Goal 12: Responsible Consumption and Productions

Justification:

This project focuses on Sustainable Development Goal 11 which is Sustainable Cities and Communities. SDG 11 emphasizes equal access to public spaces, transportation, and essential services regardless of age or physical ability. In the tourism industry, many elderly individuals and persons with disabilities continue to face barriers such as inaccessible facilities, limited accessibility information, and difficulties in planning safe and convenient travel routes. These challenges often reduce their confidence to travel independently and prevent them from fully participating in tourism activities. Therefore, improving accessibility through digital technology is an important step towards creating a more inclusive tourism environment where everyone can enjoy equal opportunities to explore tourist destinations.

The proposed Smart Accessibility Guide for Sustainable Tourism supports SDG 11 by providing a mobile application that enables disabled and elderly tourists to plan and experience their trips more independently, safely, and conveniently. Users can create personalized profiles based on their accessibility needs, search and filter attractions according to available accessibility facilities, receive AI-powered recommendations, and navigate using routes that consider accessibility requirements. The application also provides real-time accessibility information through QR code scanning, along with emergency support features such as nearby hospital locations, emergency contacts, and location sharing to improve user safety during travel. By reducing

accessibility barriers and promoting inclusive tourism, the system encourages greater participation in tourism while helping administrators maintain accurate accessibility information for attractions. Overall, the project contributes to building more inclusive communities, improving mobility for people with disabilities and elderly users, and supporting sustainable tourism development in line with the objectives of SDG 11.

1.3 Background Study

Tourism is one of the key contributors to economic growth and social development in Malaysia. The industry generates revenue while supporting businesses such as hospitality, transportation, retail, and food services. To attract more domestic and international visitors, governments invest in infrastructure and public facilities that improve the travel experience. Tourism also promotes cultural exchange through exhibitions, festivals, and events, while creating employment opportunities and encouraging local entrepreneurship. As the tourism industry continues to grow, ensuring that tourism experiences are accessible and inclusive for all visitors has become increasingly important (Yeiha, 2019; Meier, 2024).

Accessible tourism aims to ensure that people of all abilities can participate in tourism activities independently and comfortably. This includes people with disabilities, older adults, and individuals with temporary mobility limitations. According to the World Health Organization, more than one billion people, approximately 15% of the global population, live with some form of disability. Without adequate accessibility, this large group of potential travellers faces barriers when planning and enjoying their trips, while tourism businesses also miss valuable economic opportunities. Therefore, improving accessibility benefits both society and the tourism industry by creating a more inclusive travel environment (Senglaub, 2025; Gada, 2024).

The need for accessible tourism is particularly relevant in Malaysia. According to the Department of Statistics Malaysia (DOSM), there were 805,509 registered persons with disabilities (PWDs) in 2024, representing approximately 2.4% of the national population. Physical disabilities and learning disabilities accounted for the largest proportion of registered PWDs. Although Malaysia has improved accessibility in many public spaces through facilities such as ramps, elevators, accessible parking, and tactile pathways, the level of accessibility across tourism destinations remains inconsistent. Some attractions still lack adequate accessible facilities or fail to provide clear and reliable accessibility information for visitors (Department of Statistics Malaysia [DOSM], 2024; NeoLee, 2023).

These accessibility issues directly affect the travel experiences of persons with disabilities. A recent Malaysian study found that many people with mobility impairments travel infrequently because tourist destinations are not designed to accommodate their needs. Accessibility concerns also reduce their confidence in visiting new places and discourage future travel. These findings indicate that the challenge extends beyond physical infrastructure, as travellers also require reliable and centralized information to determine whether a destination is suitable before making travel plans (Saad et al., 2025).

In response to these challenges, several existing applications have been developed to assist disabled travelers. Three notable examples were found which are WheelMate, ElderWander, and Signly. Each targeting wheelchair users, the elderly, and the deaf respectively. WheelMate, developed by Coloplast, is a crowdsourced mobile app that allows wheelchair users to locate nearby accessible toilets and parking spaces through their interactive map (Quinn, 2020). The app information relies entirely on wheelchair users to add and verify the locations themselves. This means coverage is uneven if an area has fewer active contributors. Moreover, WheelMate is useful for facility lookup, but it falls short at being narrowly scoped as it only

includes toilets and parking only, with no support for attraction discovery, route planning, and accessibility ratings that help disabled people plan their travels.

The second app ElderWander is a digital tourism platform designed to provide accessible travel information for elders. The platform aggregates information about various points of interest, allowing older adults to find locations that meet their needs, and uses a question-answer model to assist the tour planning process (Costa et al., 2025). However, the app was developed and evaluated exclusively in Portugal, meaning its attraction database is limited only in that region, with no coverage of other countries. Furthermore, it focuses solely on pre-trip planning and does not offer emergency support features leaving elderly users without emergency assistance when they are already at their destination.

The third app Signly is a mobile application developed for the deaf and hearing impaired users that delivers pre-recorded sign language videos to the user's smartphone. Visitors participating in venues would point their phone at exhibits or displays and will be shown sign language descriptions of the content (*Signly the Sign Language Interpreter in Your Pocket* | Deafax, 2025). Signly represents a creative approach to accessibility in tourist attraction, but its effectiveness hinges on venue participation, and its deployment is largely limited to the United Kingdom. This means the app is of little use to hearing impaired users traveling to destinations that are not UK, including Malaysia where no venues have adopted Signly.

These existing solutions demonstrate that while there tools out there that exist to help specific disability groups, none provide a comprehensive platform that addresses the full range of accessibility needs in a tourism setting. Therefore, there is a growing need for a centralized digital platform that provides tourism information together with accessibility-related details. Such a solution can help people with disabilities identify suitable tourist attractions, understand the accessibility facilities available, and plan their journeys more confidently. By improving access to reliable information, the proposed accessibility tourism application aims to support inclusive tourism while encouraging greater participation of persons with disabilities in Malaysia's tourism industry.

- Costa, R., Coelho, J., Lima, D., Duarte, C., & Moita, P. (2025). ElderWander: an accessible digital tourism platform for older adults. *Information Technology & Tourism*.
<https://doi.org/10.1007/s40558-025-00321-7>
- DOSM. (2025). *Department of Statistics Malaysia*. Dosm.gov.my.
<https://www.dosm.gov.my/portal-main/release-content/person-with-disability-statistics-malaysia-2024>
- Gada, T. (2024, September 12). Council Post: The Importance Of Accessibility: A Comprehensive Exploration. *Forbes*.
<https://www.forbes.com/councils/forbestechcouncil/2024/09/12/the-importance-of-accessibility-a-comprehensive-exploration/>
- Meier, K. (2024, November 12). *How Tourism Supports Economic Development*. Camoin Associates.
<https://camoinassociates.com/resources/tourism-and-economic-development/>
- Neolee (2023). *Accessible Disability Facilities in South East Asia - Enhance Inclusion and Mobility*. Neolee Rehab Supply Sdn Bhd.
https://www.neolee.com.my/bm/latestnews/nid/121304/?srsltid=AfmBOooAXwUmOjsiutrbS7Z7Zp8O1OMm4q7Uo_Zyb6E5yhvfXQd6Pgi
- Quinn, M. (2020, March 28). *Wheelmate – App To Find Wheelchair Friendly Restrooms & Parking*. United Spinal Resources.
<https://spinalcord.org/disability-products-services/wheelmate-app-to-find-wheelchair-friendly-restrooms-parking/>
- Saad, M., Goh, A.-A., Ahmad, S. Y., & AbuKhalifeh, A. N. (2025). *Tourism Experiences of Persons with Disabilities in Malaysia: Voices of mobility-impaired people* [Review of *Tourism Experiences of Persons with Disabilities in Malaysia: Voices of mobility-impaired people*].
https://www.researchgate.net/publication/391511329_Tourism_Experiences_of_Persons_with_Disabilities_in_Malaysia_Voices_of_mobility-impaired_people
- Senglaub, C. (2023, July 19). *What is Accessibility in Tourism?* Travel Alliance Partnership.
<https://travelalliancepartnership.com/accessibility-in-tourism/>
- Signly the Sign Language Interpreter in Your Pocket | Deafax*. (2025). Deafax.org.
<https://www.deafax.org/news/signly-the-sign-language-interpreter-in-your-pocket/>
- Yehia, Y. (2019). *The Importance of Tourism on Economies and Businesses*. Globaledge.msu.edu.
<https://globaledge.msu.edu/blog/post/55748/the-importance-of-tourism-on-economies-a>

1.4 Project Significance

This project will contribute to :

a. Enhancing Accessibility and Inclusivity in Tourism

This system aims to provide reliable and personalized accessibility information for people with disabilities, older adults, and users with special accessibility needs, enabling them to participate more actively and independently in tourism activities. This helps create a more inclusive tourism environment that meets the needs of diverse users.

b. Supporting Sustainable Tourism Development

Utilizing smart technologies and accessibility-centric services, this system promotes an inclusive, safe, accessible, and sustainable tourism environment. This helps develop sustainable tourism practices and improve the quality of life for local communities

c. Encouraging Community Participation and Improving Accessibility Management in Tourism

Through the implementation of QR code-based crowdsourcing and user feedback mechanisms, tourists and local communities can contribute up-to-date accessibility information. This collaborative approach helps tourism authorities and attraction operators identify accessibility issues, improve facilities, and raise awareness of inclusive tourism practices, ultimately benefiting society and supporting sustainable tourism development.

2. TEAM STRUCTURE & WORK PLAN

2.1 Team Member Roles and Responsibilities

ROLE AND RESPONSIBILITIES	TEAM MEMBER
Project Manager	Oe Khye Jin
Requirement Lead	Mah Jinn Huei
Design Lead	Mah Chun How
Coding/Development	Lo Tze Khang
Programmer	1. Lo Tze Khang
Testing Lead	Jason Chew Jia Jun

2.2 Work Plan

PROJECT ACTIVITIES	TIME / DURATION
Project Planning and Proposal Preparation	15/06/2026 - 28/06/2026
Identify system requirements based on the proposal	29/06/2026 - 05/07/2026
Construct Use Case Diagrams	06/07/2026 - 09/07/2026
Develop Use Case Descriptions	10/07/2026 - 12/07/2026
Complete the requirement specification document	13/07/2026 - 19/07/2026
System Design	20/07/2026 - 02/08/2026
Mobile Application Development	03/08/2026 - 30/08/2026
System Testing and Refinement	31/08/2026 - 06/09/2026
Project Presentation and Submission	07/09/2026 - 18/09/2026

3. PROJECT FUNCTIONALITIES

3.1 Technology Stack

The proposed system will be developed as a mobile application using Flutter to provide cross-platform solutions for Android and IOS devices which can lead to reducing development time and maintenance effort. Since tourists require convenient access to accessibility information, navigation assistance, emergency support, and QR code scanning while travelling. A mobile platform enables users to access these services anytime and anywhere using their smartphones, making it more suitable than a web-based or standalone system. Furthermore, mobile devices have built-in features such as GPS, camera, internet connectivity, and notifications, which are essential for route planning, location sharing, emergency assistance, and on-site QR code scanning.

Firebase will be utilized to manage user accounts, application data, and uploaded images because they provide scalable cloud-based services with real-time data synchronization. OpenStreetMap and GPS services will be integrated to support location-based functions such as attraction discovery, route navigation, and emergency location sharing, while QR code scanning library will enable users to retrieve accessibility information and submit condition reports on-site. In addition, GitHub will be used for version control and collaborative development among team members.

3.2 Key Modules and Functionalities

MODULE	FUNCTIONALITIES
Tourism Support and Emergency Module	- Emergency contact management - Allow users to add and manage emergency contacts for safety purposes. - Location sharing - Allow users to share their current location. - Accessibility assistance information - Provide information about nearby accessible facilities and support services for disabled travellers.
Accessible Attraction Directory & Recommendation Module	- Attraction Search and Accessibility Filtering - Allow users to search for tourist attractions using keywords, location, or categories, and filter - Attraction Details - Provide detailed information about attractions, including accessibility facilities, operating hours, fees, and descriptions.

	3. User Feedback and Ratings - Allow users to submit reviews and ratings based on their experiences. 4. Budget Planning - Allow users to estimate and plan travel expenses based on attraction fees and transportation costs. 5. Preference-Based Recommendation - Generate personalized attraction recommendations using user preferences, favorites, and history.
Smart Route & Navigation Module	1. Accessible route planning - Provides optimized travel routes based on users' accessibility needs, such as wheelchair-friendly paths and safer walking paths for elderly users. 2. Public transportation information - Displays nearby bus stops, train stations, and available public transport routes to help users plan their journey conveniently. 3. Parking information - Shows available parking areas near tourist attractions, including accessible parking spaces for disabled users. 4. Travel time estimation - Calculates and provides estimated travel time based on selected routes and transportation modes such as walking, driving, or public transport.

User Management Module

1. User registration
 - Allow the users to create the account using their own personal information such as name, password, email address, contact, and emergency contacts.
2. Profile management
 - Enable the users to view or edit their personal information, such as profile picture, account settings, managing emergency contact and reset their passwords.
3. Accessibility preferences
 - Allows users to select accessibility requirements such as wheelchair access, elderly-friendly facilities, visual impairment support, and hearing impairment support during the registration phase to receive personalized recommendations, and they can edit the preferences later in the settings section.
4. Managing favourites destinations
 - Allows the user to mark the specific tourist attraction as favourite so that they can include the location as a prioritized location in their trips.
5. Manage attraction listings
 - Allow the admin to add, edit, update and remove the tourist attraction alongside their accessibility information.
6. Accessibility reports moderation
 - Allow the admin to review and verify the flagged reports that are submitted by the user regarding the facilities and the accessibility information before updating the information of the specific tourist spots.
7. Manage users accounts
 - Admin are given the authority to suspend users, reactivating accounts and verify the trusted contributor. User suspension will be given when there are users violating the platform's policy. Users are able to appeal to lift the ban. Users with the most points will be given

	the trusted contributor role by the admin. 8. View platform analytics - Displays statistical information such as active users, most visited attractions, most frequently reported accessibility issues, and contributor reward points.
On-site QR Module	1. QR scanning at location - Users can scan a QR code placed at a tourist attraction using their smartphone and camera to retrieve accessibility information for that location. 2. Real time accessibility status display - After scanning users can get information of current accessibility status of the place 3. Condition reporting with photo attachment - User has option to submit a report after viewing the accessibility status, describing the condition and attach photos as evidence (report broken ramp or closed OKU toilet) 4. Scan-based point rewarding - Award points on scan, to encourage participation and continuous contribution 5. User visit history logging - Records and display all location a user has scanned with details like date, time and accessibility status, allowing users to track past contributions 6. Negative report flagging and reverification - Flags a location for admin review when it receives repeated negative reports. After a few days will prompt a selected number of users who scan that location to verify.